1.6 Exercises

September 25, 2023

1

The complex conjugate of $a - bi$ is $a + bi$.

2

A complex number is defined to be of the form $a + bi$ where a and b are real numbers. The imaginary unit i is defined to be $\sqrt{-1}$. Well, if $b = 0$, then by the zero product property, $bi = 0$, and thus, the complex number $a + bi$ would simplify to a, which was assumed to be a real number. Thus, the answer to the question is no.

3

$x^2 = -28$
$\rightarrow x = \sqrt{-28} = \pm 2i\sqrt{7}$

4

$r^2 = -624$
$\rightarrow r = \sqrt{-624} = \pm 4i\sqrt{39}$

5

$z^2 + 8 = 4$
$\rightarrow z = \sqrt{-4} = \pm 2i$

6

$s^2 - 22 = -112$
$\rightarrow s = \sqrt{-90} = \pm 3i\sqrt{10}$

7

$2x^2 + 31 = 9$
$\rightarrow x = \sqrt{-11} = \pm i\sqrt{11}$

8

$9 - 4y^2 = 57$
$\rightarrow y = \sqrt{-12} = \pm 2i\sqrt{3}$

9

$6t^2 + 5 = 2t^2 + 1$
$\rightarrow t = \sqrt{-1} = \pm i$

10

$3p^2 + 7 = -9p^2 + 4$
$\rightarrow 12p^2 = -3$
$\rightarrow p = \sqrt{-15} = \pm i\sqrt{15}$

11

$-5(n-3)^2 = 10$
$\rightarrow (n-3)^2 = -2$
$\rightarrow n = 3 \pm i\sqrt{2}$

12

$(6-3i) + (5+4i)$
$\rightarrow = 11 + i$

13

$(9+8i) + (8-9i)$
$\rightarrow = 17 - i$

14

$(-2-6i) - (4-6i)$
$\rightarrow = -6$

15

$(-1+i) - (7-5i)$
$\rightarrow = -8 + 6i$

16

$(8+20i) - (-8+12i)$
$\rightarrow = 16 - 8i$

17

$(8 - 5i) - (-11 + 4i)$
$\rightarrow = 19 - 9i$

18

$(10 - 2i) + (-11 - 7i)$
$\rightarrow = -1 - 9i$

19

$(14 + 3i) + (7 + 6i)$
$\rightarrow = 21 + 9i$

20

$(-1 + 4i) + (-9 - 2i)$
$\rightarrow -10 + 2i$

21

C

22

$6i(3 + 2i) = 18i + 12i^2 = 18i - 12 = -12 + 18i$

23

$-i(4-8i) = -4i + 8i^2 = -8 - 4i$

24

$(1-i)^3 = (1-i)(1-i)(1-i) = -2i(1-i) = -2i + 2i^2 = -2 - 2i$

25

$(-2+5i)(-1+4i) = -18 - 13i$

26

$(-2i)^4 = 16i^4 = 16$

27

$(8-3i)(8+3i) = 73$

28

$\frac{7i}{8+i} = \frac{(7i)(8-i)}{(8+i)(8-i)}$
$\rightarrow = \frac{7+56i}{65}$
$\rightarrow = \frac{7}{65} + \frac{56}{65}i$

29

$\frac{6i}{3-i} = \frac{6i(3+i)}{(3-i)(3+i)}$
$\rightarrow = \frac{-6+18i}{10}$
$\rightarrow = -\frac{3}{5} + \frac{9}{5}i$

30

$\frac{-2-5i}{3i} = \frac{(-2-5i)-3i}{3i(-3i)}$
$\rightarrow = \frac{-15+6i}{9}$
$\rightarrow = -\frac{5}{3} = \frac{2}{3}i$

31

$\frac{4+9i}{12i} = \frac{(4+9i)-12i}{12i(-12i)}$
$\rightarrow = \frac{108-48i}{-144}$
$\rightarrow = -\frac{3}{4} + 13i$

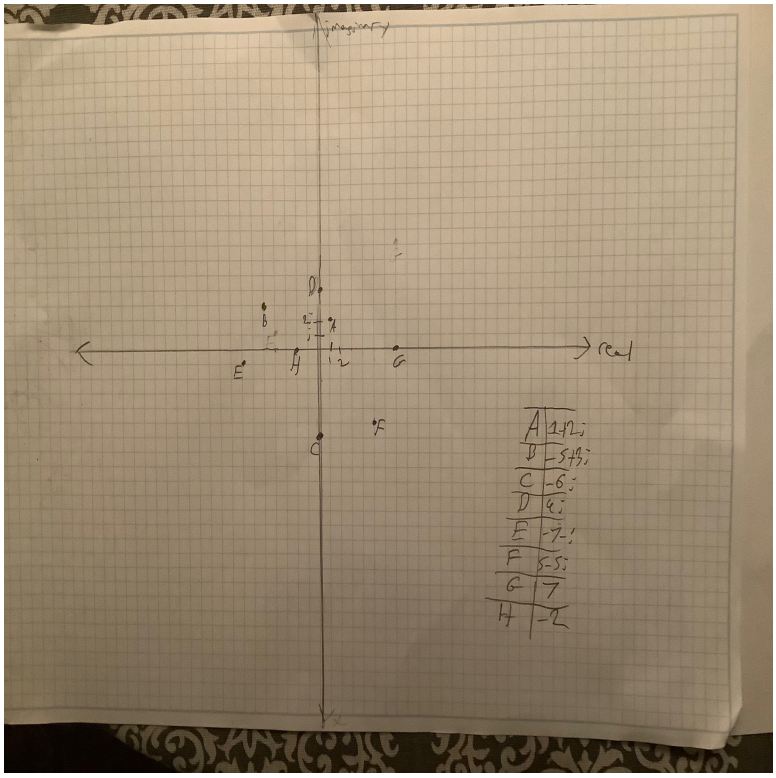
32

$\frac{7+4i}{2-3i} = \frac{(7+4i)(2+3i)}{(2-3i)(2+3i)}$
$\rightarrow = \frac{2+29i}{13}$
$\rightarrow = \frac{2}{13} + \frac{29}{13}i$

33

$\frac{-1-6i}{5+9i} = \frac{(-1-6i)(5-9i)}{(5+9i)(5-9i)}$
$\rightarrow = \frac{-59-21i}{106}$
$\rightarrow = -\frac{59}{34} - \frac{21}{106}i$

34



42

$4 + 3i$
$\rightarrow \sqrt{4^2 + 3^2} = 5$

43

$-3 + 10i$
$\rightarrow \sqrt{(-3)^2 + 10^2} = \sqrt{109}$

44

$10 - 7i$
$\rightarrow \sqrt{10^2 + (-7)^2} = \sqrt{149}$

45

$-1 - 6i$
$\rightarrow \sqrt{(-1)^2 + (-6)^2} = \sqrt{37}$

46

$-8i = 0 - 8i$
$\rightarrow \sqrt{0^2 + (-8)^2} = \sqrt{64} = 8$

47

$4i = 0 + 4i$
$\rightarrow \sqrt{0^2 + 4^2} = \sqrt{16} = 4$

48

$-4 + i$
$\rightarrow \sqrt{(-4)^2 + 1^2} = \sqrt{17}$

49

$7 + 7i$
$\rightarrow \sqrt{7^2 + 7^2} = \sqrt{98}$

50

$9 + 12i$
$\rightarrow \sqrt{9^2 + 12^2} = \sqrt{225} = 15$, so the answer is B.

51

$-8 - (3 + 2i) - (9 - 4i)$
$\rightarrow = -20 + i$

52

$(3 + 2i) + (5 - i) + 6i$
$\rightarrow = 8 + 7i$

53

$5i(3 + 2i)(8 + 3i)$
$\rightarrow = (-10 + 15i)(8 + 3i) = -125 + 90i$

54

$(1 - 9i)(1 - 4i)(4 - 3i)$
$\rightarrow = (-35 - 13i)(4 - 3i) = -179 + 53i$

55

$\frac{(5-2i)+(5+3i)}{(1+i)-(2-4i)} = \frac{10+i}{-1+5i}$
$\rightarrow = \frac{(10+i)(-1-5i)}{(-1+5i)(-1-5i)}$
$\rightarrow = \frac{-5-51i}{26} = -\frac{5}{26} - \frac{51}{26}i$

56

$\frac{(10+4i)-(3-2i)}{(6-7i)(1-2i)} = \frac{7+6i}{-8-19i}$
$\rightarrow = \frac{(7+6i)(-8+19i)}{(-8-19i)(-8+19i)}$
$\rightarrow = \frac{-17+66i}{425} = -\frac{1}{25} + 66425i$

57

This answer is not in standard form. $-2i^2 = 2$, and thus the answer can be corrected to $6 + 7i$.

58

This answer forgets to put parenthesis around -3 when it is being squared. Corrected, the answer is:
$\sqrt{2^2 + (-3)^2} = \sqrt{13}$

59

(a): Additive inverse: $-2 - i$ Multiplicative inverse: $z = \frac{1}{2+i}$
$\rightarrow = \frac{1(2-i)}{(2+i)(2-i)}$
$\rightarrow = \frac{2-i}{5} = \frac{2}{5} - \frac{1}{5}i$
(b): Additive inverse: $-5 + i$ Multiplicative inverse: $z = \frac{1}{5-i}$
$\rightarrow = \frac{1(5+i)}{(5-i)(5+i)}$
$\rightarrow = \frac{5+i}{26} = \frac{5}{26} + \frac{1}{26}i$
(c): Additive inverse: $1 - 3i$ Multiplicative inverse: $z = \frac{1}{-1+3i}$
$\rightarrow = \frac{1(-1-3i)}{(-1+3i)(-1-3i)}$
$\rightarrow = \frac{-1-3i}{10} = -\frac{1}{10} - \frac{3}{10}i$

60

$2+6i$ and $-4-6i$ are two such examples. These numbers are related such that, if we let $b_1 = 6$ and $b_2 = -6$ in the above two imaginary numbers, $b_1 + b_2 = 0$.

61

$$\frac{a+bi}{c+di}$$
$$\rightarrow= \frac{(a+bi)(c-di)}{(c+di)(c-di)}$$
$$\rightarrow= \frac{(ac+bd)+(bc-ad)i}{c^2+d^2}$$
$$\rightarrow= \frac{ac+bd}{c^2+d^2} + \frac{bc-ad}{c^2+d^2}i$$

62

$$\frac{a-bi}{c-di}$$
$$\rightarrow= \frac{(a-bi)(c+di)}{(c-di)(c+di)}$$
$$\rightarrow= \frac{(ac+bd)+(ad-bc)i}{c^2+d^2}$$
$$\rightarrow= \frac{ac+bd}{c^2+d^2} + \frac{ad-bc}{c^2+d^2}i$$

63

$$\frac{a+bi}{c-di}$$
$$\rightarrow= \frac{(a+bi)(c+di)}{(c-di)(c+di)}$$
$$\rightarrow= \frac{(ac-bd)+(ad+bc)i}{c^2+d^2}$$
$$\rightarrow= \frac{ac-bd}{c^2+d^2} + \frac{ad+bc}{c^2+d^2}i$$

64

$$\frac{a-bi}{c+di}$$
$$\rightarrow= \frac{(a-bi)(c-di)}{(c+di)(c-di)}$$
$$\rightarrow= \frac{(ac-bd)+(-ad-bc)i}{c^2+d^2}$$
$$\rightarrow= \frac{ac-bd}{c^2+d^2} + \frac{-ad-bc}{c^2+d^2}i$$

65

$Impendance = 4 + 6i - 9i = 4 - 3i$

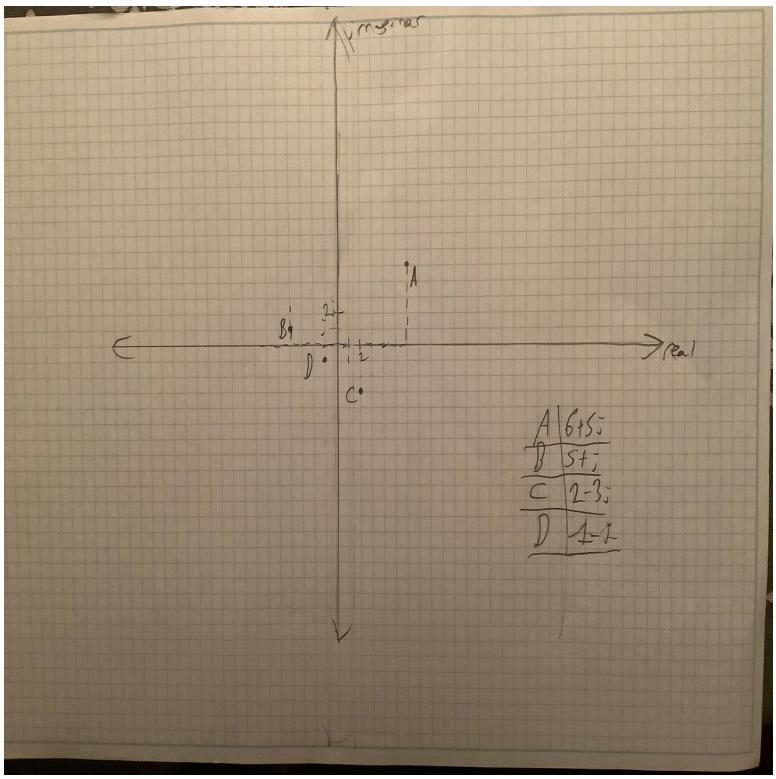
66

$Impendance = 14 + 7i - 8i = 14 - i$

67

$Impendance = 12 + 8i - 10i - 6i = 12 - 8i$

68



i^1	i^2	i^3	i^4	i^5	i^6	i^7	i^8
i	-1	$-i$	1	i	-1	$-i$	1

69

The pattern that can be concluded from this table is that the series of infinite powers of i alternates from i to -1 to $-i$ to i to -1, and then the pattern restarts. The next four powers of i are $i,\ -1,\ -i,\ 1$.

70

The communtative property of multiplication states that $a \times b = b \times a$.
$(4 - 2i)(-3 + i) = -12 + 4i + 6i - 2i^2 = -10 + 10i$
$(-3 + i)(4 - 2i) = -12 + 6i + 4i - 2i^2 = -10 + 10i$

71

The associative property of multiplication states that $(a \times b) \times c = a \times (b \times c)$.
$((7 - 2i)i)(-1 + i) = (2 + 7i)(-1 + i) = -2 + 2i - 7i + 7i^2 = -9 - 5i$
$(7 - 2i)(i(-1 + i)) = (7 - 2i)(-1 - i) = -7 - 7i + 2i + 2i^2 = -9 - 5i$

72

The distributive property states that $a(b + c) = ab + ac$.
$-2i((3 + 2i) + (5 + i)) = -2i(8 + 3i) = 6 - 16i$
$-2i(3 + 2i) - 2i(5 + i) = (4 - 6i) + (2 - 10i) = 6 - 16i$

73

The commutative property of addition states that $a + b = b + a$.
$(a + bi) + (c + di) = ac + (b + d)i$
$(c + di) + (a + bi) = ac + (b + d)i$

74

$\sqrt{-4} \times \sqrt{-25} = 2i \times 5i = -10$ while $\sqrt{100} = 10$. This suggests the fact that $\sqrt{a} \times \sqrt{b} \neq \sqrt{ab}$ when a and b are negative numbers.

75

(a): $Z = \frac{(4+5i)(7-3i)}{(4+5i)+(7-3i)} = \frac{43+23i}{11+2i} = \frac{(43+23i)(11-2i)}{(11+2i)(11-2i)}$
$\rightarrow = \frac{519+167i}{125} = \frac{519}{125} + \frac{167}{125}i$
(b): $Z = \frac{(6+8i)(10-11i)}{(6+8i)+(10-11i)} = \frac{148+14i}{16-3i} = \frac{(148+14i)(16+3i)}{(16-3i)(16+3i)}$
$\rightarrow = \frac{2326+668i}{265} = \frac{2326}{265} + \frac{668}{265}i$
(c: $Z = \frac{(3+i)(4-6i)}{(3+i)+(4-6i)} = \frac{18-14i}{7-5i} = \frac{(18-14i)(7+5i)}{(7-5i)(7+5i)}$
$\rightarrow = \frac{196-8}{74} = \frac{98}{37} - \frac{2}{37}i$

76

(a): $f(z_0) = i^2 + 1 + i = i$. Thus, because for every z_i, $f(z_i) = i$, the numbers stay in a finite bound, and hence, z_0 is a member of the Julia set corresponding to $z^2 + 1 + i$.
(b) (I was not sure how to verify for the next 3 that, for all the infinitely many z_i, whether $f(z_i)$ stayed in a finite bound, or whether these did not) : $f(z_0) = 1^1 + 1 + i = 2 + i$.
(c): $f(z_0) = (2i)^2 + 1 + i = 2 + i$.
(d): $f(z_0) = (2 + 3i)^2 + 1 + i = -4 + 13i$.